

SIT 218/738: Secure coding

Pass task 2.1P: STRIDE Threat modelling

Overview

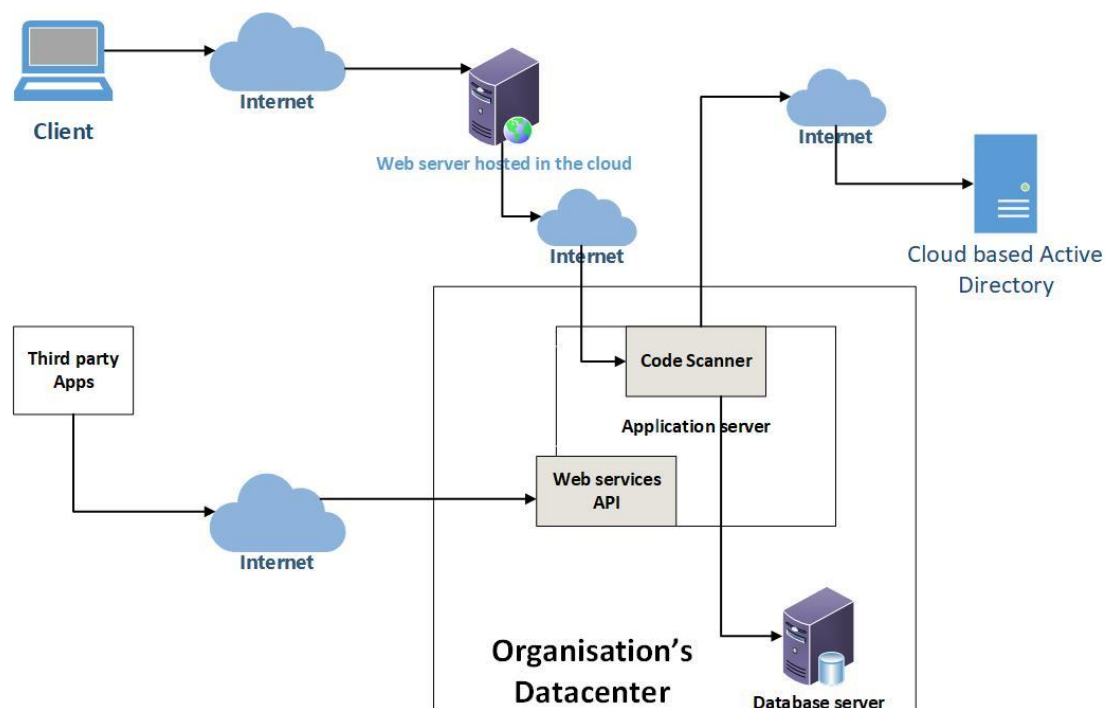
Continue your work from Task 1.2P and accomplish the STRIDE threat modelling for the following web application.

Note: Before you start working on this task, you will need to complete “the Task1.1P and 1.2P Start Activity” and study other related content in ClouDeakin unit site.

Students enrolled in SIT738 need to complete additional mandatory Tasks as mentioned below.

Application Description:

The web application is portal for performing static code analysis of software. The users of the application are software developers (who want to analyse their code), their managers, and administrators (of the web application). The users are authenticated using a cloud based active directory server which manages their identities. The software developers can upload their code to the webserver and perform a static code analysis (which is run in the application server) on the portal, view the analysis reports (scan results stored in the database) and prioritise the vulnerabilities to be addressed and resolve the fixed vulnerabilities (stored in the database). The software developers can also use their own IDEs to interact with code analyser via the web services API and perform the static code analysis. Their managers can view the vulnerabilities reported by the scan tool, view the progress made by the development team in fixing the vulnerabilities. The administrators can manage the whole web application.



Further to the DFD you completed in Task 1.2P, accomplish the STRIDE threat modeling (follow the template presented in the table below). You also need to follow threat modelling steps explained in the **Task1.1P and 1.2P Start Activity**.

For SIT218 unit: You need to select one DFD element from each category (external entity, data flow, process, and data store). Then identify any two threats for each type of DFD element in your STRIDE model, write threat scenarios and mitigation techniques for each threat. A DFD element may be associated with different threats, including spoofing, tampering, repudiation, information disclosure, denial of service, and elevation of privilege.

For SIT738 unit: You need to select one DFD element from each category (external entity, data flow, process, and data store). Then, identify any three threats for the DFD Process element; two threats (Spoofing and Repudiation) for the DFD Interactor element; three threats (Tampering, Information Disclosure, and Denial of Service) for the DFD Data Flow element; and three threats (Tampering, Information Disclosure, and Denial of Service) for the DFD Data Store element in your STRIDE model. Write threat scenarios and mitigation techniques for each threat. A DFD element may be associated with different threats, including spoofing, tampering, repudiation, information disclosure, denial of service, and elevation of privilege.

Threats	DFD element	Description of Threat events/scenarios for the element and impact of the threats	Mitigation Techniques
Spoofing			
Tampering			
Repudiation			
Information disclosure			
Denial service			
Elevation of Privileges			

Submission Requirements:

Submit one PDF file via OnTrack with required answer for this task as mentioned in the above.

Submission Due

The due for each task has been stated via its OnTrack task information dashboard.